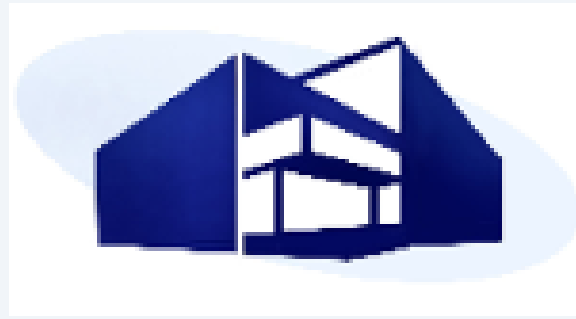
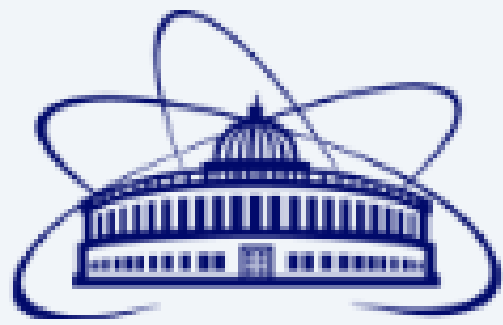
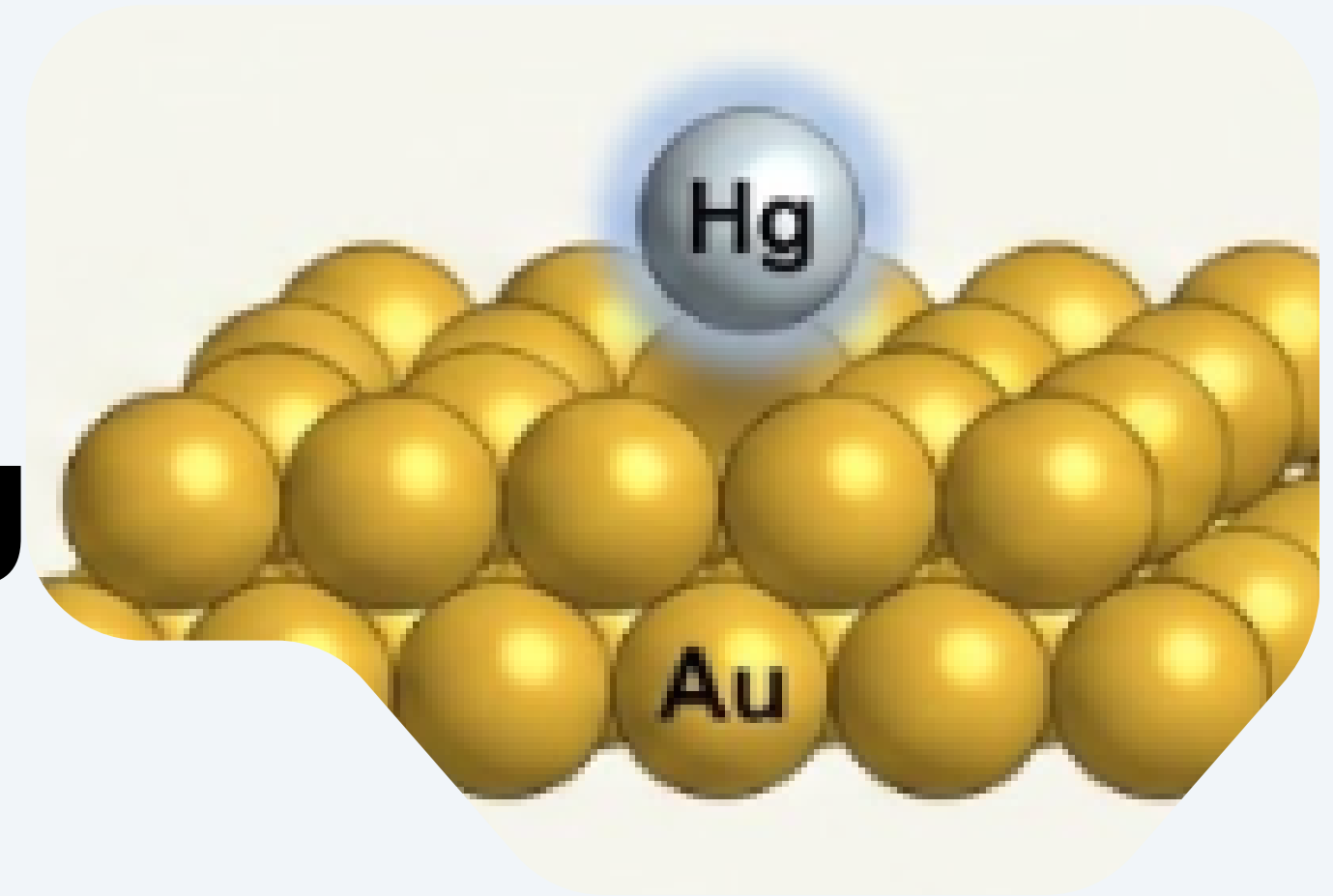




# Python driven simple atomistic modeling of mercury adsorption on gold surface



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Dr. Dipayan Sen

# Outline

01

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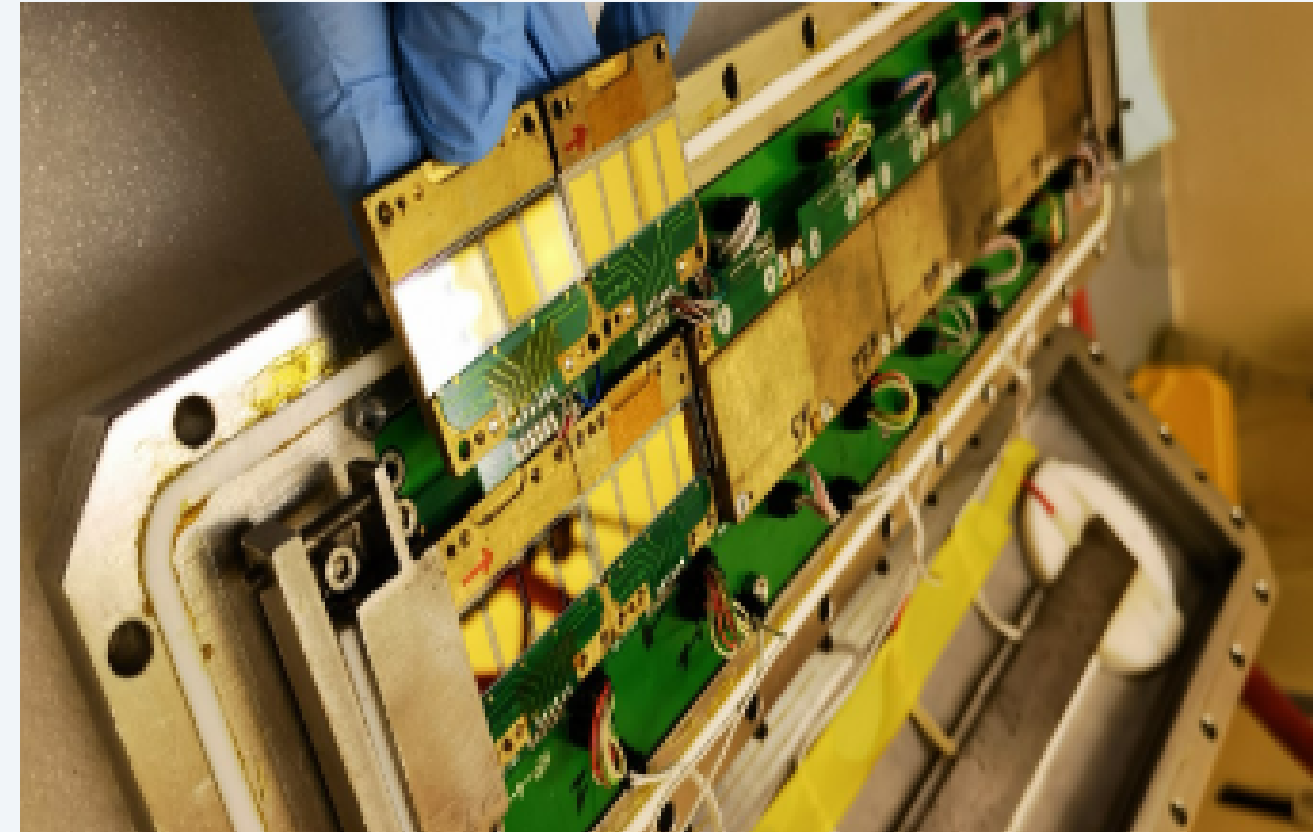
05

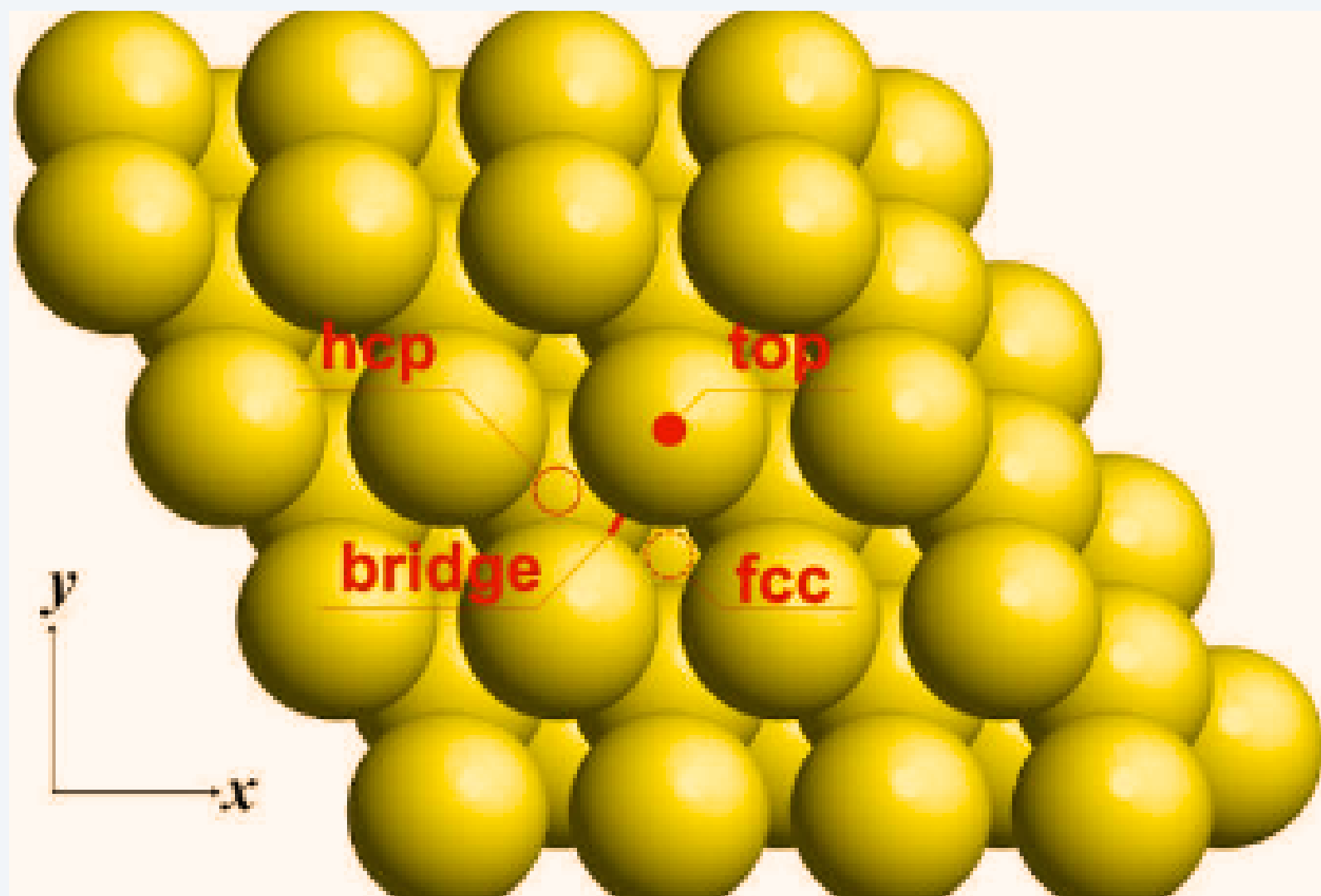
Acknowledgements



# Motivation (1)

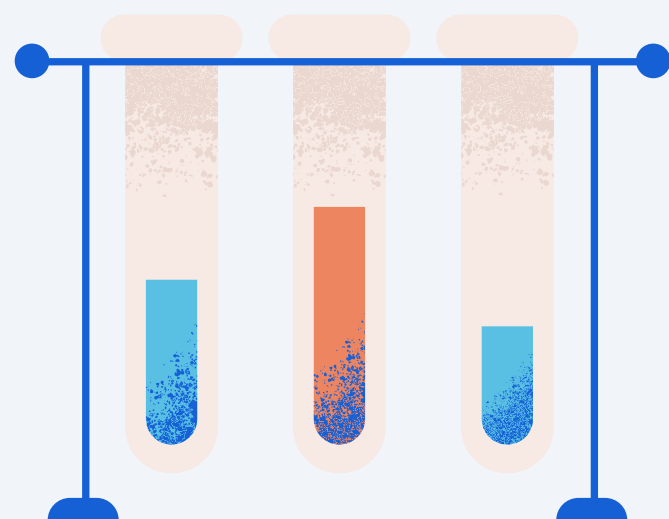
- **Flerov Laboratory of Nuclear Reactions (FLNR)** in JINR conducts experiments on the synthesis and chemical characterization of superheavy elements [1].
- Superheavy atoms are produced **only atom-by-atom** and often have **lifetimes of seconds or less** [2].
- **Gas-phase chromatography** can probe their interaction with selected surfaces.
- Experimental adsorption behavior can be used to estimate adsorption enthalpies and investigate chemical reactivity and bonding [3].
- It is important to emphasize that machine learning, as a branch of artificial intelligence, is an active area of research and development in the Russian Federation [4 - 5].
- The computational workhorse for the project is the **Atomic Simulation Environment (ASE)** with connected codes to simulate the adsorption processes of selected adsorbates and surfaces [6 - 7].



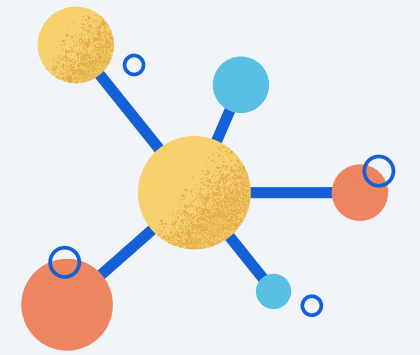


# Motivation (2): Why Hg on Au(111)?

- Mercury is the lighter Group-12 homologue of copernicium (Cn,  $Z = 112$ ), making Hg a useful reference for studying Cn-related adsorption behavior [7].
- Gold is an experimentally relevant adsorption surface in superheavy-element chemistry.
- Au(111) provides a well-defined surface for systematic atomistic modeling [2].
- A single Hg atom is studied at four high-symmetry sites [8].



# Computational Methodology (1)



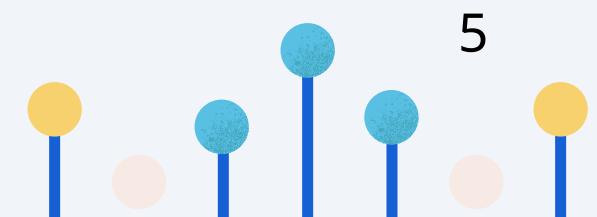
**WSL2 (Ubuntu)** – Linux environment for running atomistic simulations.

- **Miniconda3 / Conda** – separate environments for model-specific dependencies
- **Python3** – main scripting environment for MACE-based calculations.
- **ASE** – structure construction, adsorption setup, BFGS optimization, trajectories and structure I/O.
- **MLIP**- Machine Learning Interatomic Potentials: **MACE-MP-0 [9] medium** (mace-torch)-**D3 [10]** (torch-dftd)/ **CHGNet [11]-D3 / ALIGNN-FF [12]** – pretrained JARVIS-DFT force field.

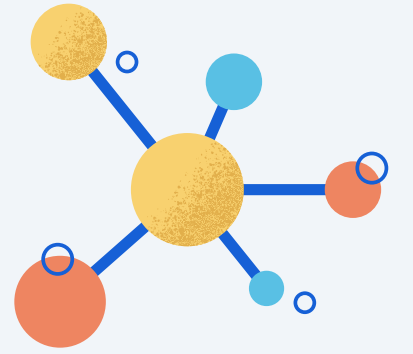
**PyTorch + CPU** – ML backend; calculations executed on CPU (**float64** where specified).

**VESTA + ASE GUI** – visualization of relaxed structures and optimization trajectories.

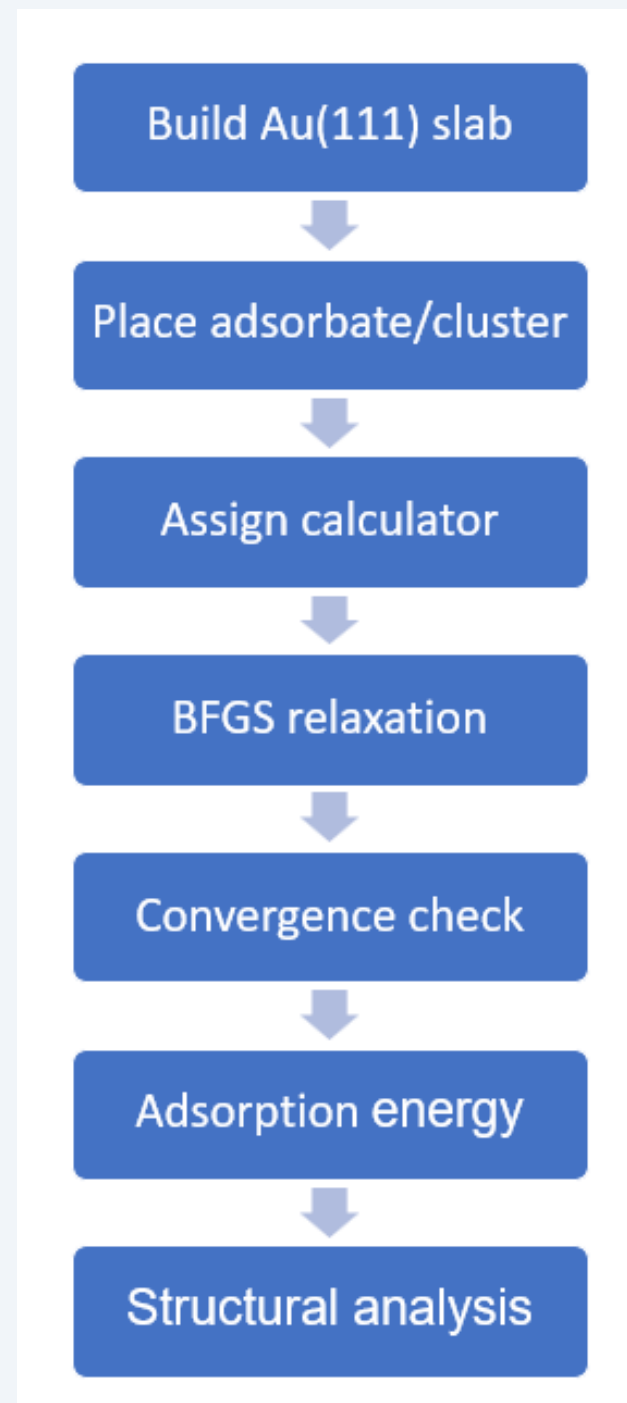
**Git / GitHub** – code version control and reproducible project organization.



# Computational Methodology (2): Workflow & Outputs



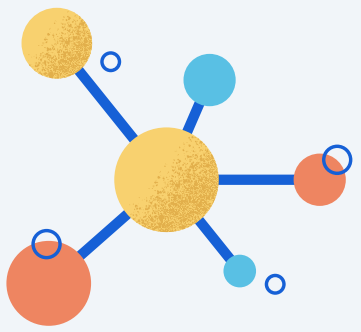
## Workflow



## Setup

1. Structure setup
  - Au(111)  $4 \times 4 \times 4$  slab
  - 64 Au atoms
  - 10 Å vacuum
  - Hg initially placed 2.0 Å above the surface
  - Sites: fcc, hcp, bridge, ontop
2. Calculator: Hg: MACE + D3 / CHGNet / ALIGNN-FF
3. Geometry optimization: BFGS optimizer/ Convergence monitored by  $f_{\max} < 0.01$  eV/Å
4. Adsorption energy:
$$E_{\text{ads}} = E_{\text{total}} - E_{\text{slab}} - E_{\text{Hg}}$$
  - $E_{\text{total}}$ : total energy of the relaxed Hg/Au(111) system
  - $E_{\text{slab}}$ : total energy of the relaxed clean Au(111) slab
  - $E_{\text{Hg}}$ : reference energy of Hg
5. Outputs
  - Python scripts (.py) – simulation workflow, parameters and model setup
  - Log files (.log) – optimization steps, energies and  $f_{\max}$ ; convergence checking
  - Trajectory files (.traj) – atomic configurations during relaxation
  - VASP structure files (.vasp) – relaxed structures for visualization in VESTA

# Python script



## Python script modifiers

- Deepseek
- ChatGPT
- Claude

```
MACE-MP-0 is used in float64 to make geometry optimization reliable.
float32 introduces ~1 meV numerical noise in the energy, which is
comparable to the fcc/hcp energy difference and would make the site
ranking unreliable.
```

```
Reference energies are recomputed at runtime so the script is fully
self-contained and reproducible.
"""
```

```
import os
import warnings
import numpy as np
from ase import Atoms
from ase.build import fcc111, add_adsorbate
from ase.optimize import BFGS
from ase.io import write
```

```
# Silence the harmless torch.load(weights_only=False) FutureWarning
warnings.filterwarnings("ignore", category=FutureWarning,
                        message=".*weights_only.*")
```

```
from mace.calculators import mace_mp
```

```
# =====
# Configuration
# =====
SLAB_SIZE   = (4, 4, 4)      # 4x4x4 Au(111) slab -> 64 Au atoms
VACUUM      = 10.0           # Angstrom
ADSORBATE   = 'Hg'           # adsorbate element
HEIGHT      = 2.0            # initial adsorbate height (Angstrom)
```

```
# Convergence: 0.01 eV/A is the standard "tight" target. In float64
# MACE forces are accurate enough to actually reach this.
```

```
FMAX        = 0.01           # force convergence (eV/Angstrom)
MAXSTEPS     = 500            # BFGS safety limit
```

```
SITES        = ['fcc', 'hcp', 'bridge', 'ontop']
```

```
# --- 0. Shared Settings ---
```

```
calc = CHGNetCalculator()
fmax_val = 0.05
```

```
def get_relaxed_energy(atoms, name):
    atoms.calc = calc
    print(f"Optimizing {name}...")
    dyn = BFGS(atoms, trajectory=f'{name}.traj', logfile=None)
    dyn.run(fmax=fmax_val)
    return atoms.get_potential_energy()
```

```
# --- 1. Isolated Hg13 Cluster ---
```

```
# Manual latticeconstant fix for Hg
```

```
hg13_iso = Icosahedron('Hg', noshells=2, latticeconstant=3.0)
```

```
hg13_iso.center(vacuum=10) # Place in a vacuum box
```

```
e_cluster = get_relaxed_energy(hg13_iso, "cluster_iso")
```

```
# --- 2. Clean Au(111) Slab ---
```

```
slab_clean = fcc111('Au', size=(4, 4, 3), vacuum=10.0)
```

```
# Fix bottom 2 layers to simulate bulk
```

```
mask = [atom.tag < 3 for atom in slab_clean]
```

```
slab_clean.set_constraint(FixAtoms(mask=mask))
```

```
e_slab = get_relaxed_energy(slab_clean, "slab_clean")
```

```
# --- 3. Combined System (Hg13 on Au) ---
```

```
# Start fresh to avoid constraint conflicts
```

```
system = fcc111('Au', size=(4, 4, 3), vacuum=10.0)
```

```
hg13_ads = Icosahedron('Hg', noshells=2, latticeconstant=3.0)
```

```
add_adsorbate(system, hg13_ads, height=2.5, position='fcc')
```

```
system.set_constraint(FixAtoms(mask=mask)) # Apply same slab constraints
```

```
e_total = get_relaxed_energy(system, "total_system")
```

```
# --- 4. Final Calculation ---
```

```
e_ads = e_total - (e_slab + e_cluster)
```

```
print("-" * 30)
```

```
print(f"E_total:    {e_total:10.4f} eV")
```

```
print(f"E_slab:     {e_slab:10.4f} eV")
```

```
print(f"E_cluster: {e_cluster:10.4f} eV")
```



# Results and Discussion

# Adsorption of Hg on Gold(111) using MACE-MP-0+D3

4 positions of Hg on Gold: fcc, hcp, bridge and top

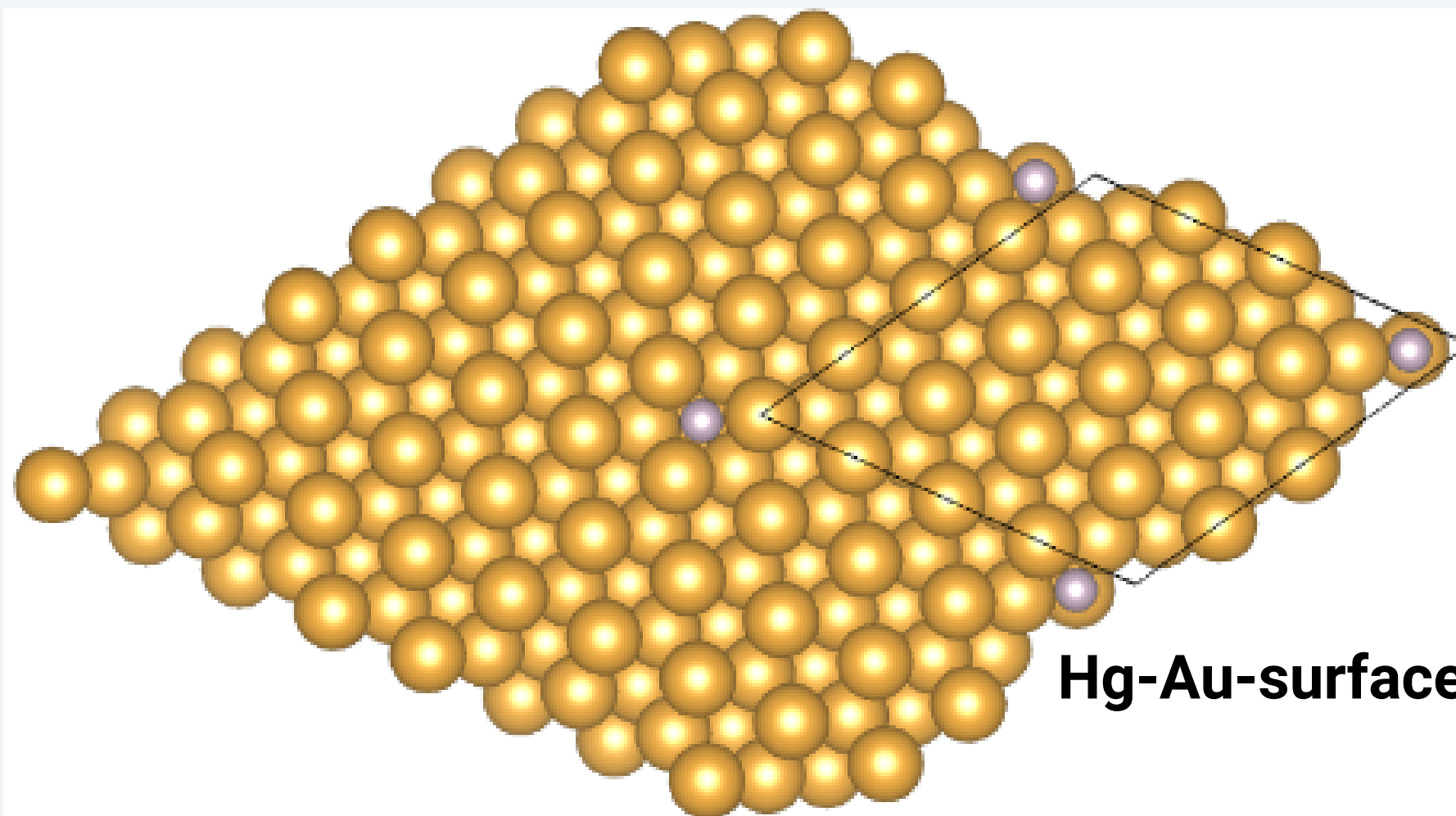
**Fcc** : -112.689 KJ/mol

Bridge : -110.113 KJ/mol

Hcp : -112.187 KJ/mol

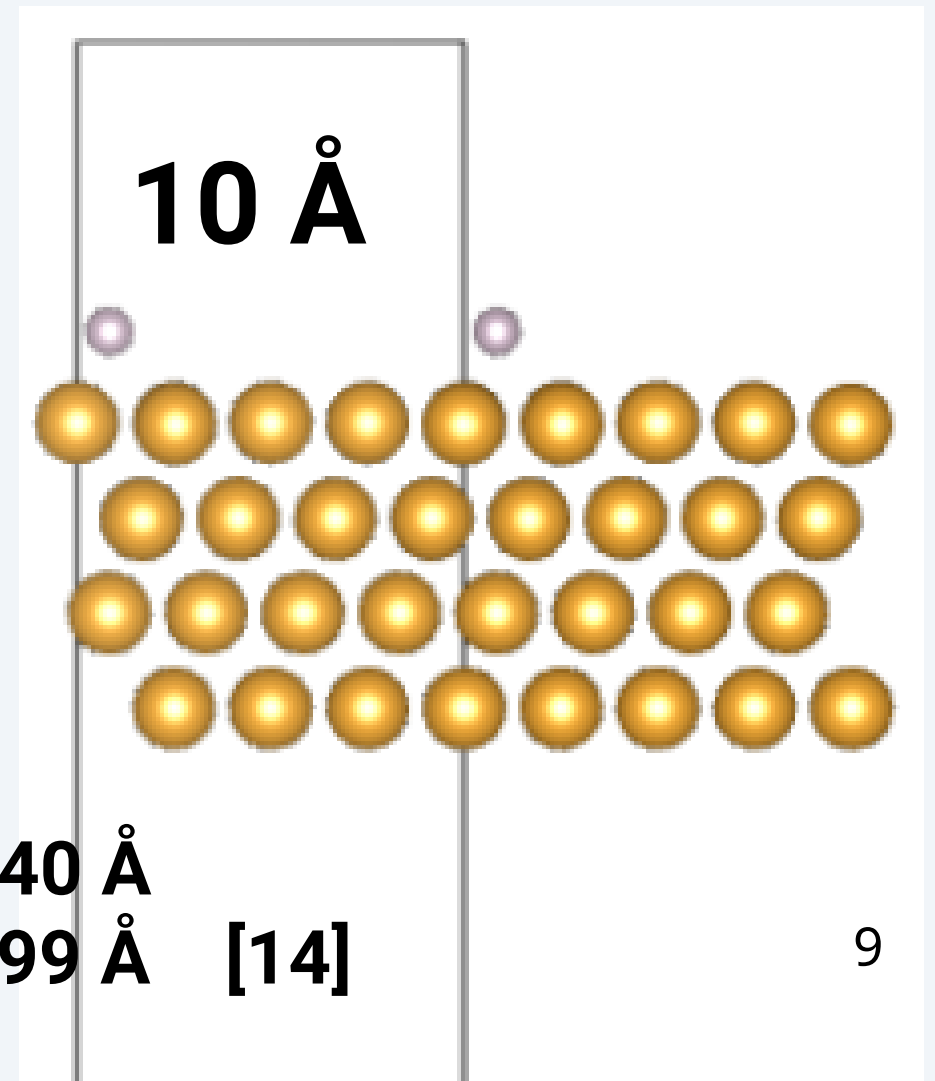
On top : -94.097 KJ/mol

**FCC  
top  
view**



Hg-Au-surface bond length : 2.940 Å

**FCC  
side  
view**



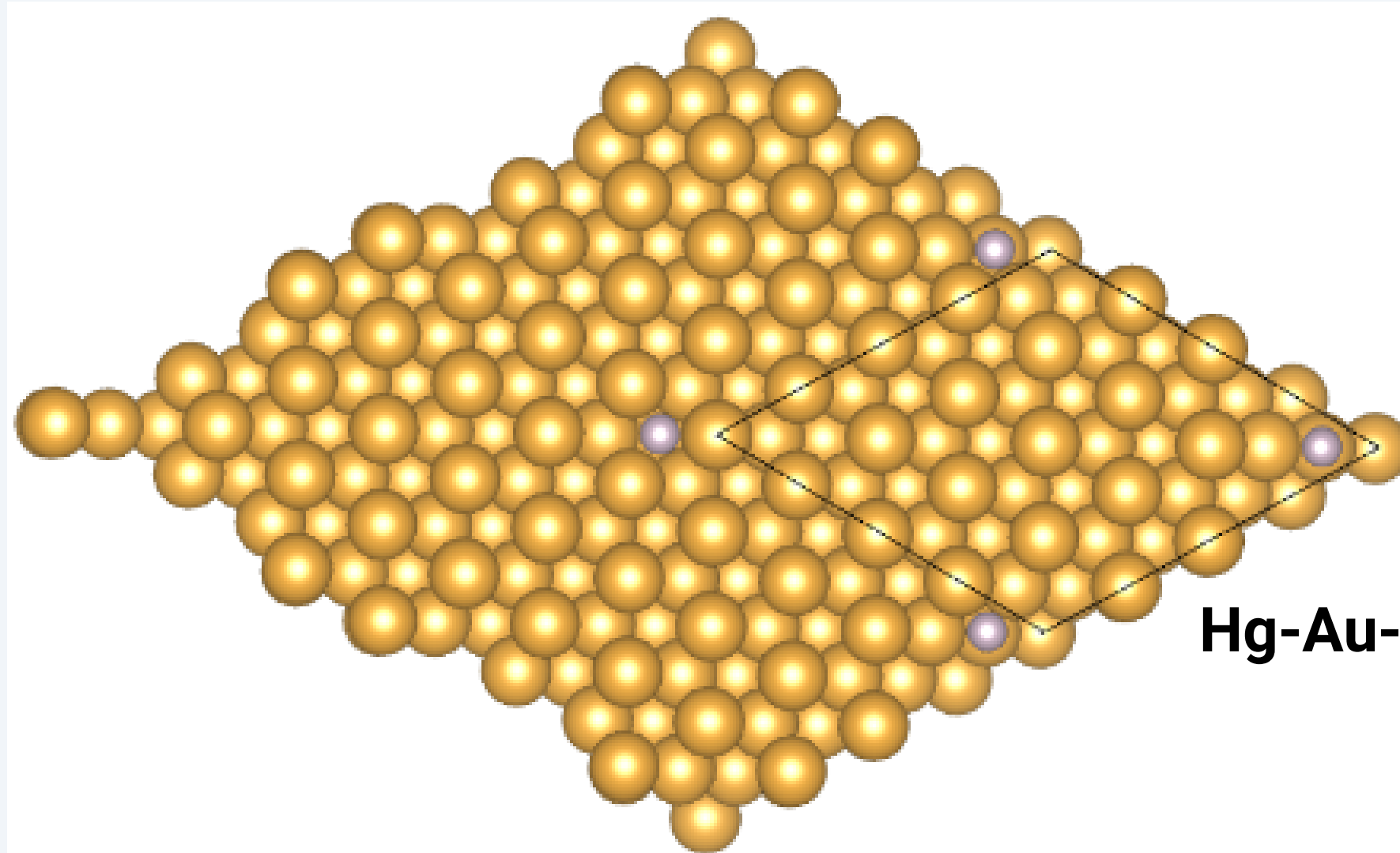
# Adsorption of Hg on Gold(111) using CHGNet

4 positions of Hg on Gold: fcc, hcp, bridge and top

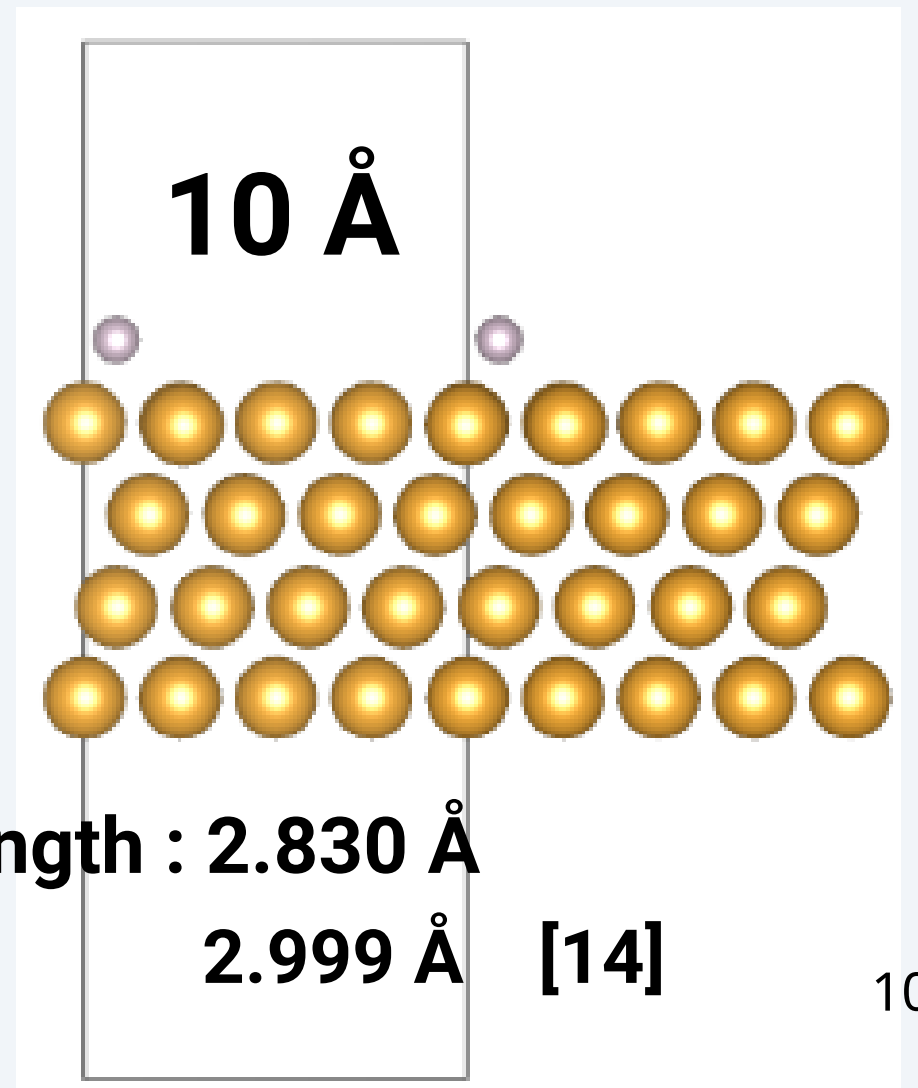
**Fcc : -105.126 KJ/mol** Bridge : -105.089 KJ/mol

Hcp : -104.659 KJ/mol On top : -83.479 KJ/mol

**FCC  
top  
view**



**FCC  
side  
view**



**Hg-Au-surface bond length : 2.830 Å**

**2.999 Å [14]**

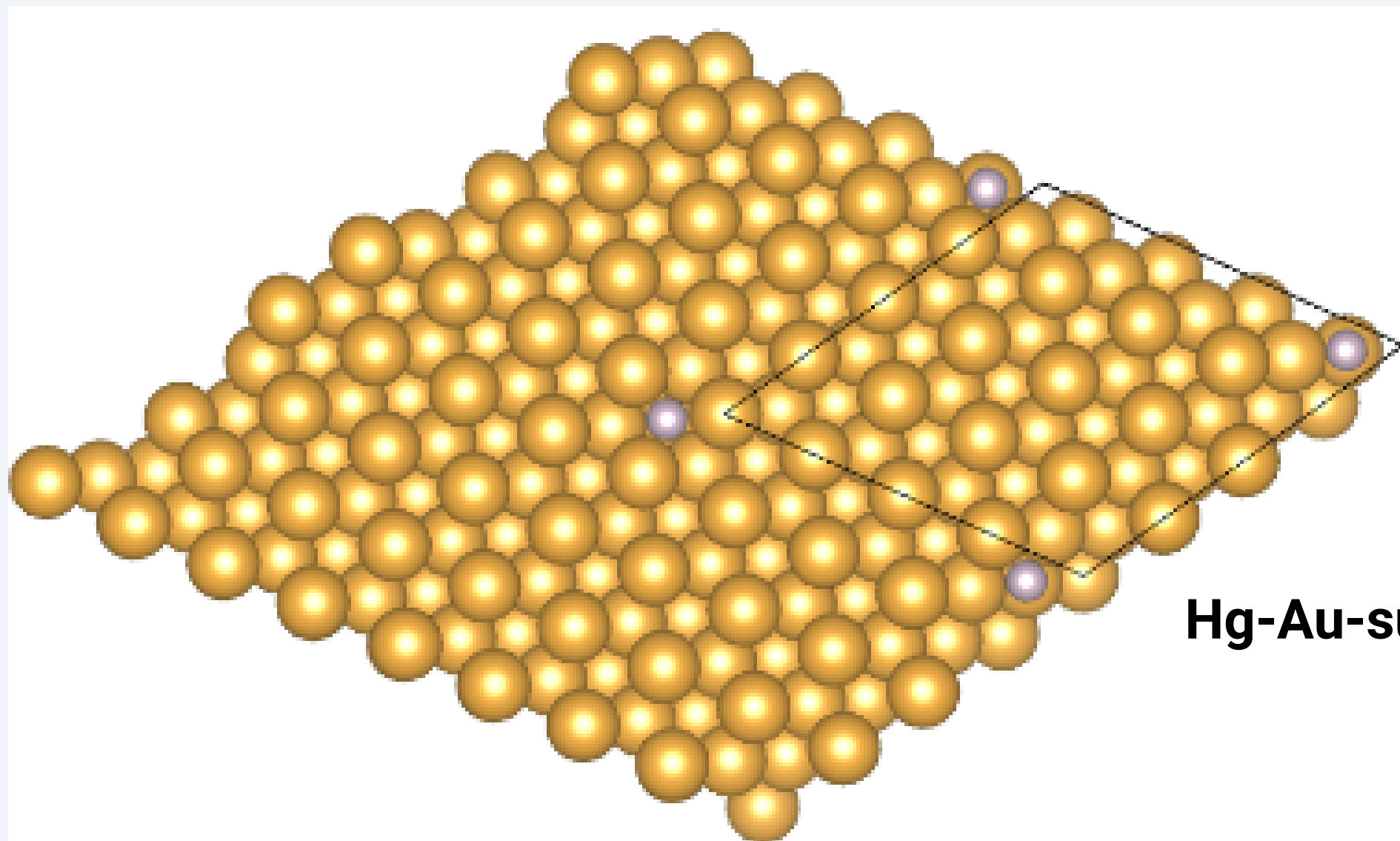
# Adsorption of Hg on Gold(111) using ALIGNN-FF

4 positions of Hg on Gold: fcc, hcp, bridge and top

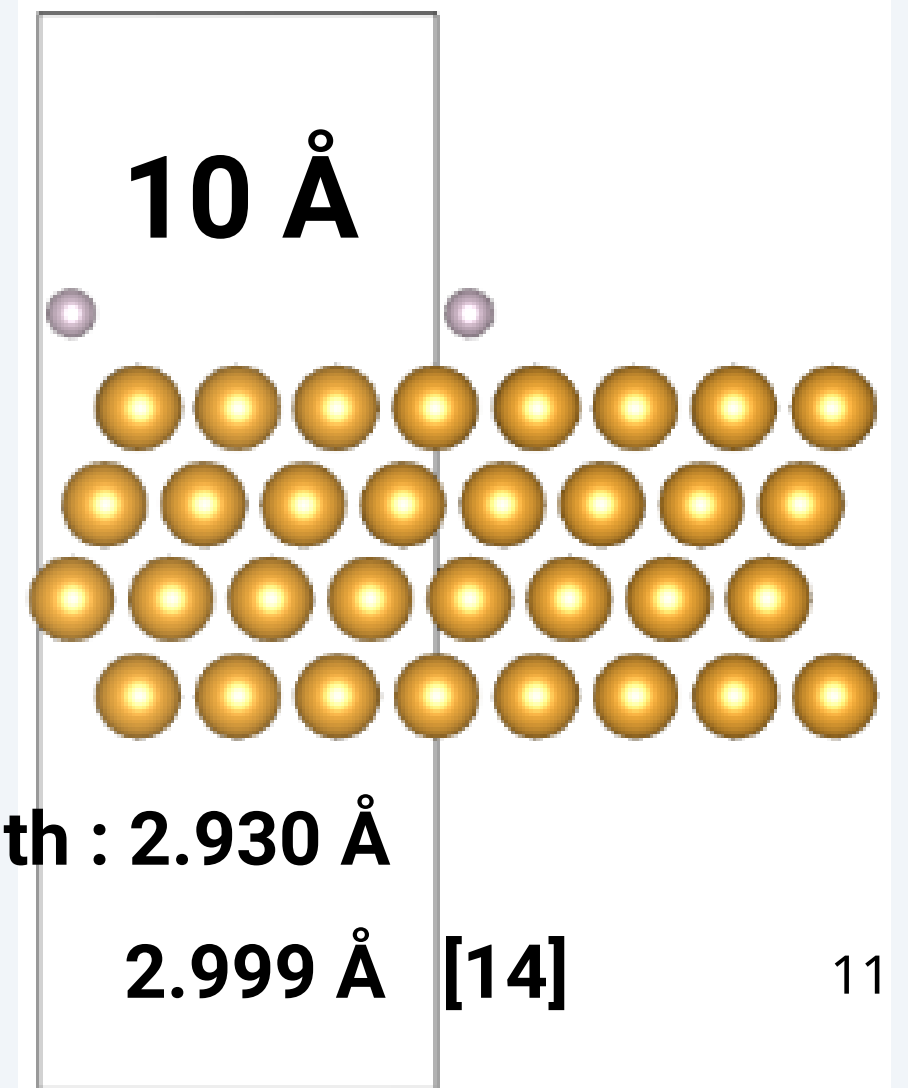
Fcc : -94.331 KJ/mol    **Bridge : -102.250 KJ/mol**

Hcp : -92.250 KJ/mol    On top : -71.796 KJ/mol

**Bridge  
top  
view**



**Bridge  
side  
view**



**Hg-Au-surface bond length : 2.930 Å**

**2.999 Å [14]**

# Summary on adsorption energies



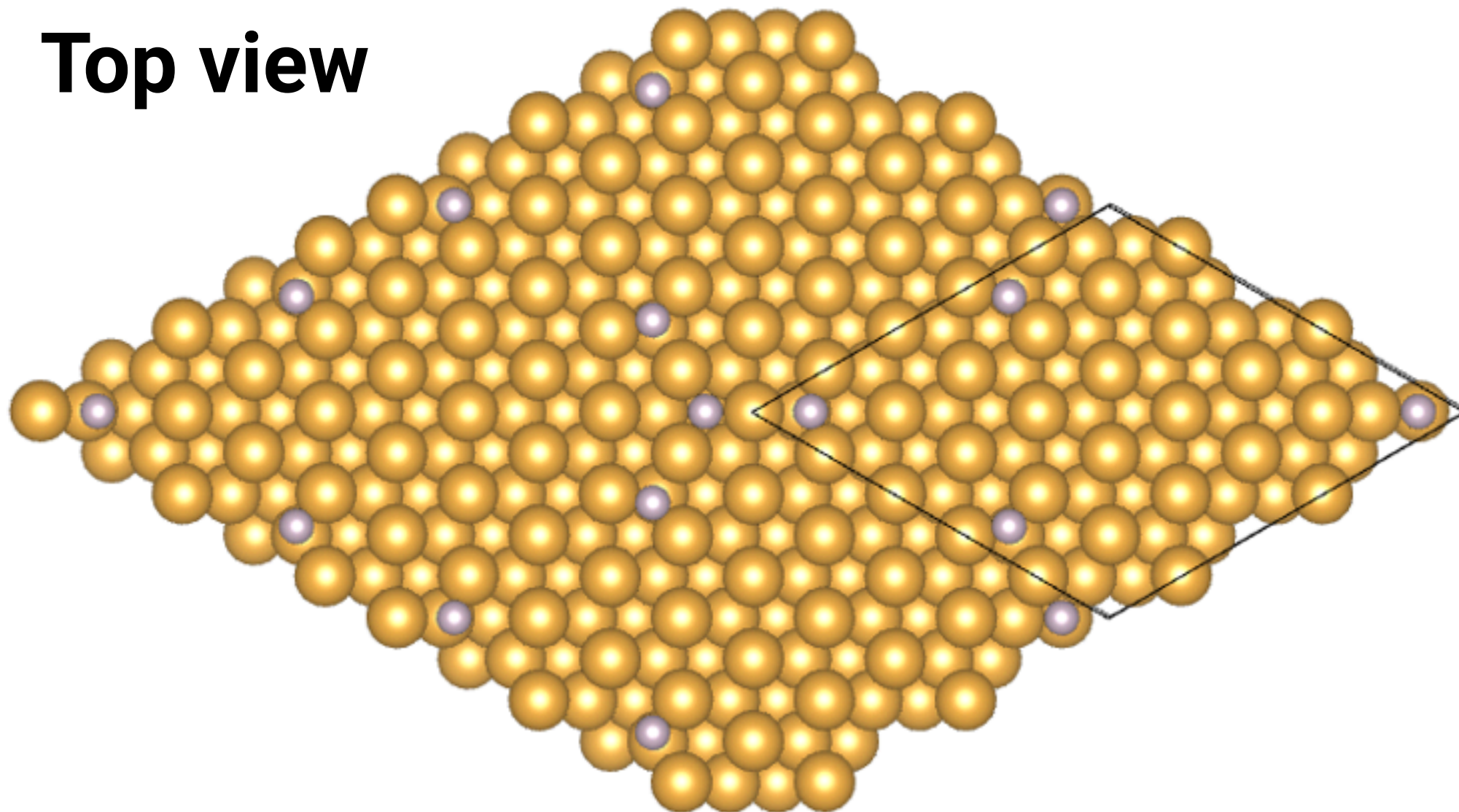
| Method       | Best site | Adsorption energy (kJ/mol) | Hg-Au bond length(Å) |
|--------------|-----------|----------------------------|----------------------|
| MACE-MP-0+D3 | FCC       | -112.689                   | 2.940                |
| CHGNet       | FCC       | -105.126                   | 2.830                |
| ALIGNN-FF    | Bridge    | -102.250                   | 2.930                |
| DFT-D3       | FCC       | -75.259 [15]               | -                    |
| Experimental | -         | -97.450 [16]               | -                    |

# Adsorption of Hg<sub>4</sub> miniCluster on Gold(111) using MACE-MP-0+D3

E<sub>adsorption</sub> energy : -386.174 KJ/mol

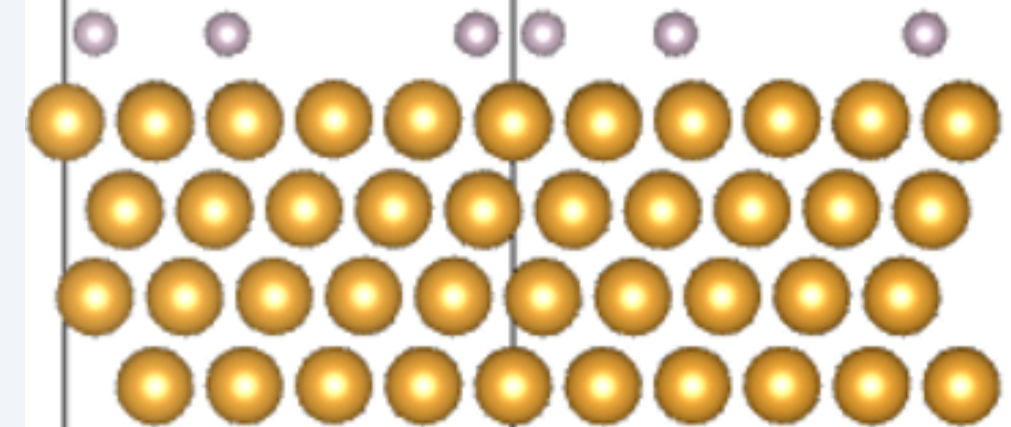
E<sub>adsorption/atom</sub> : -96.544 KJ/mol

Top view



20 Å

Side view



Hg-Au bond length : 2.999 Å

2.999 Å [14]



# Conclusion

- MACE-MP-0+D3 and CHGNet both favor the FCC site. ALIGNN-FF favors the bridge site, which makes it the outlier.
  - MACE-MP-0+D3 gives the strongest binding, stronger than CHGNet and ALIGNN-FF at the best site.
  - MACE-MP-0-D3 was used for minicluster adsorption on Au(111) since it was the best calculator for the single atom adsorption.
  - For the cluster Hg adsorption, the total adsorption energy was found to be -386.174 KJ/mol.
  - The adsorption energy per atom was found to be -96.544 KJ/mol.
- 

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# Thank You

